

ARCTO: AIoT System for Reducing Carbon Emissions using Traffic Optimization

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Abstract—In the status quo, traffic control systems operate on predetermined patterns and instructions devised from past data. While this method functions effectively for traffic under normal conditions, it becomes heavily congested and inefficient during instances of high traffic, which leads to a multitude of temporal, economic, health, and environmental harms. However, by combining traditional traffic controllers with modern technologies such as Internet of Things devices and computer vision, these issues can be effectively addressed. This research presents a novel, affordable Artificial Intelligence of Things traffic control system that enables accurate real-time vehicle detection and signal control. This work is split into two sections: (1) an AIoT physical system that can scan traffic conditions in real-time and (2) a realistic traffic simulator with a custom optimization algorithm. Combined, this research provides up to 35% greater throughput, 50% reduced waiting time, and 50% reduction in greenhouse gas emission reductions in comparison to non-optimized algorithms used in the status quo. The implementation of this work leads to various temporal, economic, environmental, and health benefits; in addition to providing comparable emission reduction as the complete replacement of all internal combustion engine vehicles with battery electric vehicles, while significantly reducing vehicle travel time, systems installation time, and cost.

Index Terms—artificial intelligence of things, vehicle emissions, adaptive traffic signal control, intelligent transport systems, deep learning

I. INTRODUCTION

As the number of vehicles in the world have steadily increased in past decades, the likelihood and severity of vehicle congestion have also increased significantly. In recent years, drivers spent nearly 99 hours a year stuck in congestion, costing \$305 billion [1]. Moreover, increased congestion has led to a multitude of additional problems, such as increased greenhouse gas (GHG) emissions from vehicles, and increased deaths from pollution. On average, between 17,000 to 20,000 deaths annually in the United States are attributable to vehicle pollution [2]. Vehicle congestion is typically worst in cities and time spent waiting at intersections account for between 20% to 50% of congestion. In the status quo, traffic control systems operate on pre-determined patterns from past data. While this method functions effectively for traffic under regular conditions, it becomes heavily congested and inefficient during instances of high traffic as the system is unable to adapt to real-time changes.

In recent years, adoptions of intelligent transportation systems (ITS) for real-time vehicle detection and traffic control

have been proposed, leveraging techniques such as adaptive traffic signal control (ATSC) and dynamic vehicle rerouting.

In this paper, a portable, affordable Artificial Intelligence of Things (AIoT) vehicle detection and traffic light control system is proposed. To the author's best knowledge, this is the first implementation of such a system to address this issue. Unlike prior works, this paper presents an end-to-end system, starting from a real-time hardware vehicle detection system that can be placed on streets, to traffic signal duration optimization, all while costing under \$1000.

This work is split into two sections: (1) an AIoT physical system that can detect the number of vehicles in each lane and direction in real-time and (2) a realistic traffic simulator with a custom traffic light optimization algorithm. Combined, the system provides up to 35% increased vehicle throughput, 50% reduction to waiting time, and 50% reduction in greenhouse gas emissions in comparison to the status quo. The implementation of this work leads to various temporal, economic, environmental, and health benefits; in addition to providing comparable emission reduction as the complete replacement of all internal combustion engine vehicles with battery electric vehicles, while significantly reducing vehicle travel time, systems installation time, and cost.

In the following section, related works on intelligent traffic control, real-time vehicle detection, and methods to reduce vehicle greenhouse gas (GHG) emissions are introduced. Section III introduces the AIoT system, including the physical hardware, vehicle detection, and lane detection algorithms. Section IV introduces the traffic optimization scheme and describes the realistic simulator devised to verify the algorithm's performance. Section V presents the results and discussion of this work and a conclusion is provided in Section VI.

II. RELATED WORKS

A. Intelligent Traffic Signal Control

In recent years, intelligent transportation systems (ITS) have been deployed around the world to address issues with traffic congestion. Among strategies to resolve traffic congestion, adaptive traffic signal control (ATSC) has emerged as a popular approach due to its relatively easier implementation and widespread applicability.

An important element of ATSC is the real-time detection of vehicles. Two common approaches used are embedding

sensors under roads and vehicle detection using cameras. In [4], Zhang et al. leverage piezoelectric sensors embedded under roads to sense pressure changes induced by traffic, while McGowen et al. [5] used pneumatic road tubes to detect vehicles. However, such embedded systems are expensive to install, maintain, and replace, often requiring the entire road to be closed for repairs [6].

An alternative to these systems, like the system proposed in this paper, is the use of cameras to detect vehicles using deep learning methods. In recent years, major advancements in computer vision have enabled fast and accurate real-time object detection with algorithms such as R-CNN [7] and You Only Look Once (YOLO) [8]. For instance, in [9], Liang et al. used a variation of the YOLOv3 algorithm [10] and a kernel correlation filter algorithm to count vehicles in real-time while Youssef et al. used a novel cascade R-CNN combined with Feature Pyramid Networks [11] to automatically count and track vehicles in [12].

B. Vehicle Greenhouse Gas Emissions

In its most recent August 2021 report, the Intergovernmental Panel on Climate Change (IPCC) reiterated the severity of the climate situation and called for more action [13]. In just this past year, countries around the world, spanning from the United States to Tunisia to South Korea have experienced record-shattering natural disasters exacerbated by the worsening climate crisis. It is evident that more work must be done to address this issue.

In just the United States, transportation account for nearly 30% of all greenhouse gas (GHG) emissions, among which 83% comes from road vehicles. The elimination of this source of GHG would significantly reduce emissions across the globe. Moreover, additional temporal, environmental, economic, and health benefits arise from reduced vehicle emissions. In the past decade, reductions in vehicle emissions have saved the United States nearly 20,000 lives every year from vehicle pollution-induced deaths [2]. In 2017 alone, the economic cost of congestion on drivers amounted to \$305 billion [1].

A common approach to address vehicle emissions has been the adoption of electric vehicles. However, electric vehicles are not entirely “green” as significant GHG are expended when mining for minerals used to build electric batteries, and for the transportation of goods.

As demonstrated by [14] in the city of Xian, traffic control can be leveraged as an effective tool to immediately reduce greenhouse gas emissions for all vehicles on the road, while longer-term approaches like replacing all non-electric vehicles with electric ones are underway.

Evidently, the implementation of an intelligent, affordable, AIoT traffic optimizer will bring a multitude of advantages spanning from climate change, to the economy, and human health.



Fig. 1. Physical Traffic Control System. a) Full system, b) Circular base located at the bottom of the system, c) Camera module located at the top of the system.

III. AIoT TRAFFIC CONTROLLER

A. Physical System

The purpose of the physical hardware system is to detect vehicles and their lanes in real-time in all four directions of the intersection. Thus, the requirements for the physical hardware system are as follows: (1) simultaneous real-time video in all 4 directions, (2) clear visibility, (3) safety, and (4) affordability. To satisfy these four requirements, the physical system shown in Fig 1 was developed and tested.

A traffic cone (Fig 1a) was used to encase the overall structure due to its light weight, large size, and universality as a warning system for drivers. This alerts drivers and pedestrians and prevents them from hitting the system. An adjustable 2m tall music stand was used to support the camera module located at the top of the system and was selected because it was extremely light, and its height was adjustable. This adjustability allowed for the system to alter its height to see over the tops of vehicles at an intersection and shrink down for easy transport and storage.

A circular base (Fig 1b) was attached to the bottom of the system to support the traffic cone. Both the iEnergy 7 battery and Intel NUC PCs were housed on the circular base to maintain a low center of gravity, making the system more stable and less likely to fall due to vibrations from passing vehicles. The base was built using 5mm thick black acrylic plates, the shapes and holes were designed using the CAD software Autodesk Fusion360, and cut using an industrial laser cutter.

Lastly, a camera module (Fig 1c) consisting of a flashing warning light and four Samsung USB cameras was attached at the top. Each USB camera was attached to each of the four sides of the module and is capable of taking real-time video of resolution 1920 x 1080 pixels and 30fps. From here, the real-time video was fed to a vehicle and lane detection algorithm to count the number of vehicles per lane and per direction. Like the circular base, the camera module was built using 5mm thick black acrylic plates, with cutouts designed with Autodesk Fusion360, and cut with a laser cutter.

B. Vehicle and Lane Detection Algorithm

A new vehicle and lane detection algorithm was devised to detect the location and number of vehicles in each lane for each direction. By detecting vehicle volume imbalances in real-time, this enables the system to alter the duration of green lights for different directions, increasing throughput while reducing waiting time and emissions. Moreover, the object detection algorithm can distinguish between emergency and non-emergency vehicles, allowing for emergency vehicles to be given signal priority and pass through the intersection without delay.

A custom vehicle detection algorithm was devised from YOLOv3-tiny [15], which was selected for its accuracy and light weight. However, due to the algorithm's high computational cost and hardware limitations to maintain low price, a smaller, simplified, custom network was devised using the Darknet C framework [16].

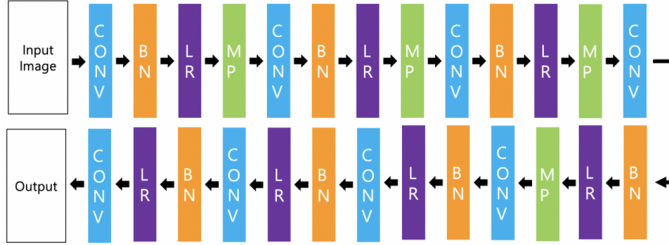


Fig. 2. Custom Vehicle Detection Algorithm Architecture. CONV represents Convolutional Layers, BN represents 2D Batch Normalization, LR represents Leaky Rectified Linear Unit (ReLU) activation functions, and MP represents 2D Max Pooling.

TABLE I

ARCHITECTURAL DIFFERENCES BETWEEN THE CUSTOM ALGORITHM AND YOLOV3-TINY.

	YOLOv3-tiny	Custom Detector	Reduction
Total # of layers	43	26	-39.53%
Total # of conv layers	13	8	-38.46%
Total # of parameters	8,669,002	726,069	-91.62%

The architecture of the custom vehicle detection algorithm is shown in Fig 2 with major reductions in the number of layers, convolutional layers, and the total number of parameters compared to YOLOv3-tiny as shown in Table II. The custom detector consists of three blocks of Convolutional, 2D Batch

Normalization, and Leaky ReLU layers, with 4 blocks of 2D MaxPooling, Convolutional, 2D Batch Normalization, and Leaky ReLU layers, followed by a fully connected convolutional layer. From the standard YOLOv3-tiny algorithm, one of the two 26x26x26 detection heads was removed because it was used for detecting small objects, which would not be necessary when detecting vehicles. Moreover, numerous convolutional layers were removed to simplify the detector to for speed and accuracy within the physical system's constraints. Despite the reductions in size and complexity of the detector, its actual overall performance was superior to that of YOLOv3-tiny, as shown in Table II.

To determine the number of vehicles per lane, the bounding box coordinates for each detected vehicle were compared to the coordinates of the detected lane markers. Lane markers were detected using a multi-step process involving canny edge detection [18], gaussian smoothing, road masking, and Hough transforms [17]. Intermediary steps and results are shown in Fig 3.



Fig. 3. Lane detection. (a) Canny Edge Detection [18], (b) Full detection with Hough Transform [17]

IV. TRAFFIC CONTROL OPTIMIZATION

A. Simulation Environment

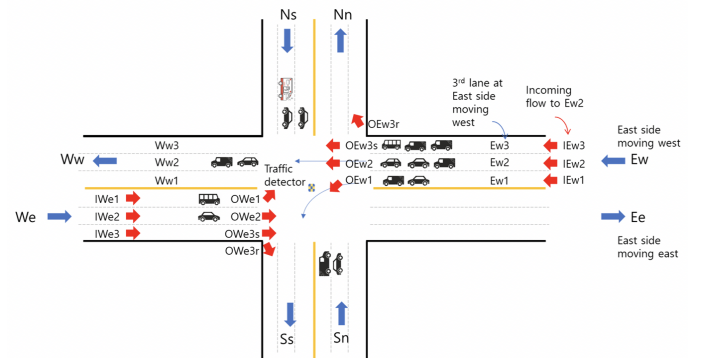


Fig. 4. Sample four-way intersection showing notation. Lanes on the east side of the intersection heading west are labeled Ew (East-to-west), with the numbers 1, 2, 3 describing the lane number.

As real traffic lights could not be connected to and altered to verify the effectiveness of the proposed system, a custom traffic simulation was devised depicting flow in a four-way intersection like Fig 4. Each direction is three lanes wide, with vehicles in the first lane only turning left, vehicles in the second lane only going straight, and vehicles in the third lane can either turning right or going straight. In the simulation,

vehicles drive on the right-hand side of the road. Also, all vehicle instances were coded to have immediate responses to the changes to the color of the traffic light and all vehicles are coded to have a minimum separation distance of 0.5 m.

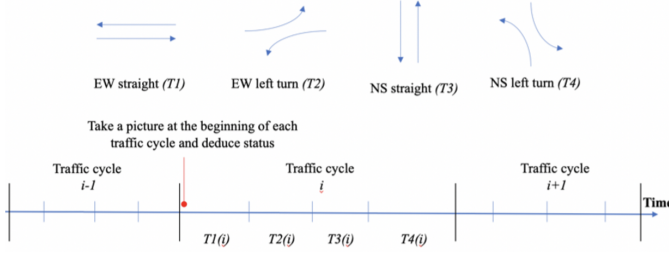


Fig. 5. Traffic signal order used in the simulation. The four traffic states used in the simulation are East-West straight, East-West left turn, North-South straight, and North-South left turn, mirroring real-world traffic light patterns.

Fig 5 shows the traffic signal cycle in the simulated intersection. $T1$, $T2$, $T3$, and $T4$ each represent the duration of time of a specific traffic signal state. Each cycle of $T1$ to $T4$ lasts 120 seconds in total, with a yellow light period for 5 seconds at the end of each state. $T1$ and $T3$ are assigned a minimum of 10 seconds to allow time for pedestrians to cross the intersection. The objective function of the simulation is to maximize the throughput of vehicles by controlling the duration of each of the signal patterns $T1$ to $T4$.

At the start of every traffic signal cycle, the physical AIoT system takes videos of all 4 directions and determines the number of vehicles in each lane and direction. Then, the respective values are stored in two state matrices E and N , which log the number of emergency and non-emergency vehicles in each lane respectively. Emergency vehicles are independently detected as they are assigned higher priority — if they are detected, the traffic signals will default to green in the emergency vehicle's direction, allowing for it to pass through the intersection first, before resuming normal traffic light patterns.

B. Traffic Optimization Algorithm

A custom analytical algorithm was devised to optimize the duration of $T1$ to $T4$ to maximize throughput, minimize waiting time, and minimize vehicle emissions.

The algorithm is described in (1), where Q_i is the number of predicted vehicles in a specific direction at the beginning of cycle i and $\frac{\partial Q_i}{\partial t}$ represents the change in the number of vehicles between cycles i and $i - 1$.

$$Q_i = Q_{i-1} + \frac{\partial Q_i}{\partial t} \quad (1)$$

As $\frac{\partial Q}{\partial t}$ requires knowledge about the future traffic cycle, (2) is used to estimate the total number of vehicles in the intersection during the subsequent cycle by approximating from the most recently completed cycle. In (2), G_{i-1} is the number of vehicles generated into the intersection during cycle $i - 1$ and L_{i-1} is the number of vehicles leaving the intersection during cycle $i - 1$.

$$Q_i \approx Q_{i-1} + G_{i-1} - L_{i-1} \quad (2)$$

With the Q_i value, the duration of the signals for $T1$ to $T4$ is determined proportionally. The values of $T1$ to $T4$ can be calculated using (3), (4), (5), (6), and (7).

$$\begin{aligned} T1 &= \frac{100 \times \alpha}{\alpha + \beta + \gamma + \delta}, & T2 &= \frac{100 \times \beta}{\alpha + \beta + \gamma + \delta}, \\ T3 &= \frac{100 \times \gamma}{\alpha + \beta + \gamma + \delta}, & T4 &= \frac{100 \times \delta}{\alpha + \beta + \gamma + \delta} \end{aligned} \quad (3)$$

$$\alpha = \frac{(Ew2 + l3_{bias} \times Ew3 + We2 + l3_{bias} \times We3)}{4} + \frac{(IEw2 + l3_{bias} \times IEw3 + IW e2 + l3_{bias} \times IW e3)}{4} \quad (4)$$

$$\beta = \frac{(Ew1 + We1) + (IEw1 + IW e1)}{2} \quad (5)$$

$$\gamma = \frac{(Ns2 + l3_{bias} \times Ns3 + Sn2 + l3_{bias} \times Sn3)}{4} + \frac{(INs2 + l3_{bias} \times INs3 + ISn2 + l3_{bias} \times ISn3)}{4} \quad (6)$$

$$\delta = \frac{(Ns1 + Sn1) + (INs1 + ISn1)}{2} \quad (7)$$

In (4) and (6), $l3_{bias}$ is the proportion of vehicles in lane 3 that are traveling straight instead of turning right. In (4), $IEw2$ is the number of vehicles entering lane $Ew2$ and can be calculated according to (8). $Ew2_{prev}$ is the number of vehicles in lane $Ew2$ in the previous traffic cycle and $OEw2_{prev}$ is the number of vehicles from that previous cycle that successfully crossed the intersection. Since $IEw2$ is a quantity that describes the number of vehicles that will enter the lane in the following traffic cycle, data from the previous cycle is used to predict this quantity.

$$IEw2 = Ew2 - Ew2_{prev} + OEw2_{prev} \quad (8)$$

(9) presents the equivalent equation but for lane 3. Similar to (8), $Ew3_{prev}$ is the number of vehicles in the lane during the previous traffic cycle. Since vehicles in lane 3 can either take a right turn or travel straight, the $OEw3_{prev}$ value is split into the sum of two variables $OEw3s_{prev}$ and $OEw3r_{prev}$, which are the number of vehicles that successfully went straight and took a right turn respectively at the intersection.

$$IEw3 = Ew3 - Ew3_{prev} + OEw3s_{prev} + OEw3r_{prev} \quad (9)$$

Swapping out Ew for We , Ns or Sn in (8) and (9) lead to identical results for different lanes in different directions.

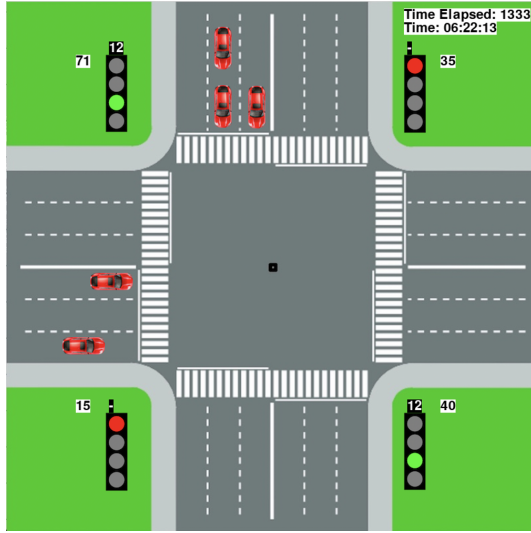


Fig. 6. Traffic simulator devised using Pygame.

C. Traffic Simulator

Instead of using a commercialized simulation, a custom simulator was developed using Pygame to allow for flexibility to expand to future research projects. The simulator receives $T1$ to $T4$ from Fig 5 and the N and E state matrices as input. Then, it sets the duration of each traffic signal and generates the required number of vehicles for each lane. The simulation operates at an accelerated pace, and at the conclusion of the simulated hour, the system records the total throughput, waiting time, and emissions of vehicles in the intersection along with the overall time elapsed.

V. RESULTS AND DISCUSSION

The performance of both the physical AIoT system and traffic control optimizer are evaluated to verify the improved capabilities of the proposed system.

TABLE II
COMPARISON BETWEEN VEHICLE DETECTION RESULTS OF YOLOV3-TINY AND CUSTOM DETECTOR.

mAP@0.5	YOLOv3-tiny	Custom Detector
All classes	0.774	0.79
Car class	0.619	0.681
Emergency class	0.93	0.899

For the physical AIoT system, Table II shows the difference in detection results between YOLOv3-tiny and the custom detector. The custom detector outperformed YOLOv3-tiny, despite being a greatly simplified version with less parameters, by having an all class mAP@0.5 of 0.79, exceeding YOLOv3-tiny's 0.774; and having a car class mAP@0.5 of 0.681, exceeding YOLOv3-tiny's 0.619. However, YOLOv3-tiny did perform better on the emergency vehicle class, with a slightly higher score of 0.93 over the custom detector's 0.899. The custom detector's improved performance may be attributable

to the removal of a detection head used for detecting smaller objects.

The throughput, waiting time, and emissions results from the simulation are shown in Fig 7, Fig 8, and 9 respectively.

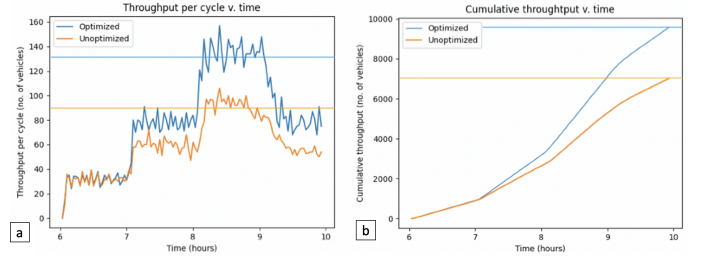


Fig. 7. Results showing the intersection throughput from the traffic simulation, with the results from the unoptimized algorithm in orange and optimized algorithm in blue. The graphs (a) shows the total vehicle throughput per 2 minute cycle and (b) shows the total vehicle throughput throughout the 4 hour simulation.

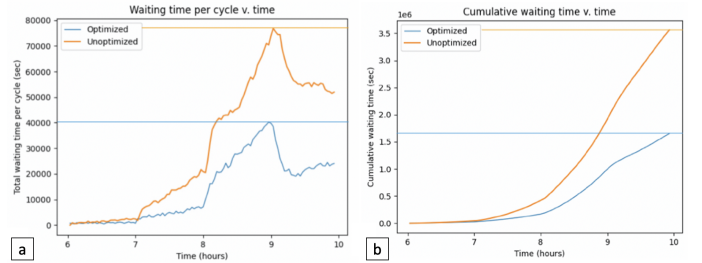


Fig. 8. Results showing the vehicle waiting time from the traffic simulation, with the results from the unoptimized algorithm in orange and optimized algorithm in blue. The graphs (a) shows the total vehicle waiting time per 2 minute cycle and (b) shows the total vehicle waiting time throughout the 4 hour simulation.

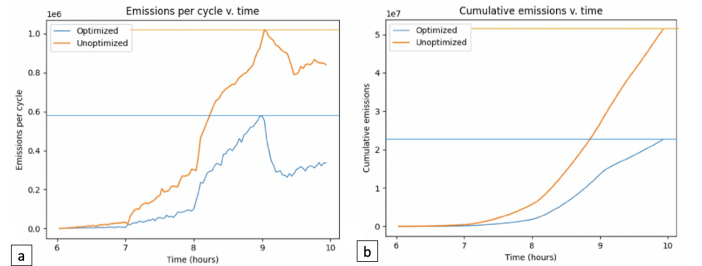


Fig. 9. Results showing the vehicle emissions generated throughput from the traffic simulation, with the results from the unoptimized algorithm in orange and optimized algorithm in blue. The graphs (a) shows the total vehicle emissions per 2 minute cycle and (b) shows the total vehicle emissions throughout the 4 hour simulation.

Traffic conditions in the simulator were devised over a 4-hour period from 6-10AM, with 1-hour long intervals of low, medium, and high traffic density. 6-7AM was at low density with 480 cars/h throughout the intersection, 7-8AM and 9-10AM were both at medium density with 1440 cars/h while 8-9AM, normally when rush hour traffic is the highest, was at high density of 4320 cars/h.

Observing Fig 7, 8, and 9, the non-optimized and optimized algorithms provide comparable results under low traffic conditions, but their difference grows under medium conditions and differs greatly during the high condition. This phenomenon is attributable to the insufficient vehicle density in the low traffic condition, such that vehicles don't "pile up" after each traffic cycle. However, a noticeable gap, both in throughput per cycle and overall throughput begins to occur from the medium density condition at 7AM, the results of which last till the end of the simulation.

Overall, the proposed system provides up to 35% greater throughput, a 50% reduction in waiting time, and 50% reduction in greenhouse gas emissions in comparison to the status quo.

Taking a closer look at the environmental impacts of this work, the current leading approach to reducing vehicle emissions has been the adoption of battery electric vehicles (BEVs). However, BEVs themselves are not entirely pollution free, as their lifecycle emission are around half of an internal combustion-engine vehicle, comparable to the reductions resulting from this work. [19] But, unlike the adoption of electric vehicles, the adoption of efficient traffic controllers like this work saves significantly on installation time, and cost. Thus, for large cities like New York where heavy congestion is frequent, it makes more sense to invest in and adopt the work presented in this paper. Not to mention, this approach also significantly reduces the waiting time for all drivers, a chronic issue that the adoption of electric vehicles does not address.

VI. CONCLUSION

In this paper, a novel, affordable, deep learning AIoT traffic control system was presented and tested. Compared to baseline systems used in the status quo, the proposed system provided 35% greater throughput, 50% reduced waiting time, and 50% reduced vehicle emissions while costing below \$1000. The implementation of this work leads to various temporal, economic, environmental, and health benefits; in addition to providing comparable emission reduction as the full adoption electric vehicles while significantly reducing waiting time. This paper acts as a feasibility study that provides a functioning prototype and demonstrate the effectiveness of the solution via a realistic simulation.

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